```
import pandas as pd
import numpy as np
import nltk
import matplotlib.pyplot as plt
import seaborn as sns

from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from nltk.tokenize import word_tokenize
from nltk.corpus import wordnet

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report, confu
```

```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
True
```

```
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
True
```

```
data=pd.read_csv('IMDB Dataset.csv')
```

```
import re

def preprocess(text):
    text = text.lower()
    text = re.sub(r'<.*?>', '', text)
    text = re.sub(r'[^a-zA-Z]', ' ', text)
    text = re.sub(r'\s+', ' ', text).strip()
    return text

data['clean_text'] = data['review'].apply(preprocess)
data.head()
```

	review	sentiment	clean_text
0	One of the other reviewers has mentioned that ...	positive	one of the other reviewers has mentioned that ...
1	A wonderful little production. The...	positive	a wonderful little production the filming tech...
2	I thought this was a wonderful way to spend ti...	positive	i thought this was a wonderful way to spend ti...
3	Basically there's a family where a little boy ...	negative	basically there s a family where a little boy ...
4	Petter Mattei's "Love in the Time of Money" is...	positive	petter mattei s love in the time of money is a...

Next steps: Generate code with data New interactive sheet

```
vectorizer=TfidfVectorizer(max_features=5000)
X=vectorizer.fit_transform(data['clean_text'])
y=data['sentiment']
```

```
X_train,X_test,y_train,y_test=train_test_split(
    X,y,test_size=0.2,random_state=42
)
```

```
model=MultinomialNB()
model.fit(X_train,y_train)

```

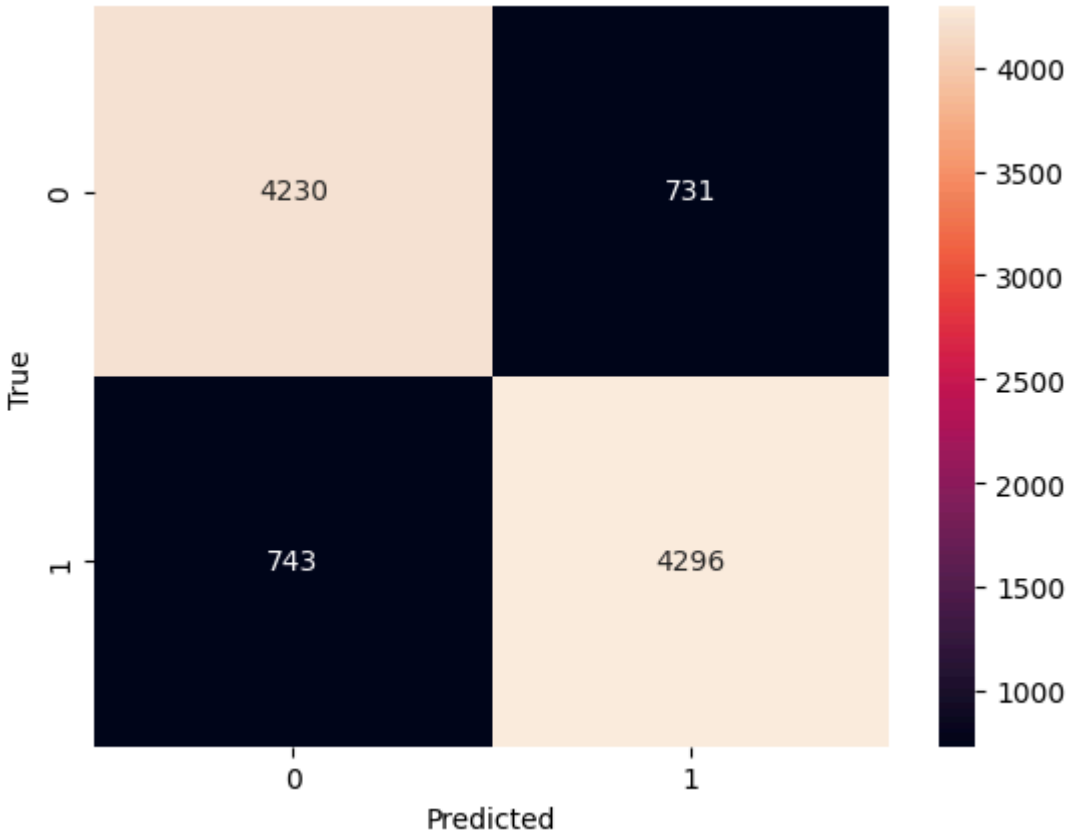
```
▾ MultinomialNB
MultinomialNB()
```

```
y_pred=model.predict(X_test)
```

```
print("Accuracy",accuracy_score(y_test,y_pred))
print("Classification Report",classification_report(y_test,y_pred))

```

```
Accuracy 0.8526
Classification Report              precision    recall  f1-score   support

    negative       0.85      0.85      0.85      4961
    positive       0.85      0.85      0.85      5039

    accuracy                           0.85     10000
   macro avg       0.85      0.85      0.85     10000
weighted avg       0.85      0.85      0.85     10000
```

```
cm=confusion_matrix(y_test,y_pred)
sns.heatmap(cm,annot=True,fmt='d')
plt.xlabel('Predicted')
plt.ylabel('True')
plt.show()
```




```
text=["This movie was fantastic and inspiring"]
clean=preprocess(text[0])
vector=vectorizer.transform([clean])
prediction=model.predict(vector)
print("Predicted Score:",prediction[0])
```

```
Predicted Score: positive
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.ensemble import RandomForestClassifier, GradientBoostingCla
```

```
from sklearn.linear_model import LogisticRegression
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all_models={
    "Naive Bayes":MultinomialNB(),
    "Logistic Regression":LogisticRegression(max_iter=200),
    "SVM":LinearSVC(),
    "Random Forest":RandomForestClassifier(),
    "Gradient Boosting":GradientBoostingClassifier()}
```
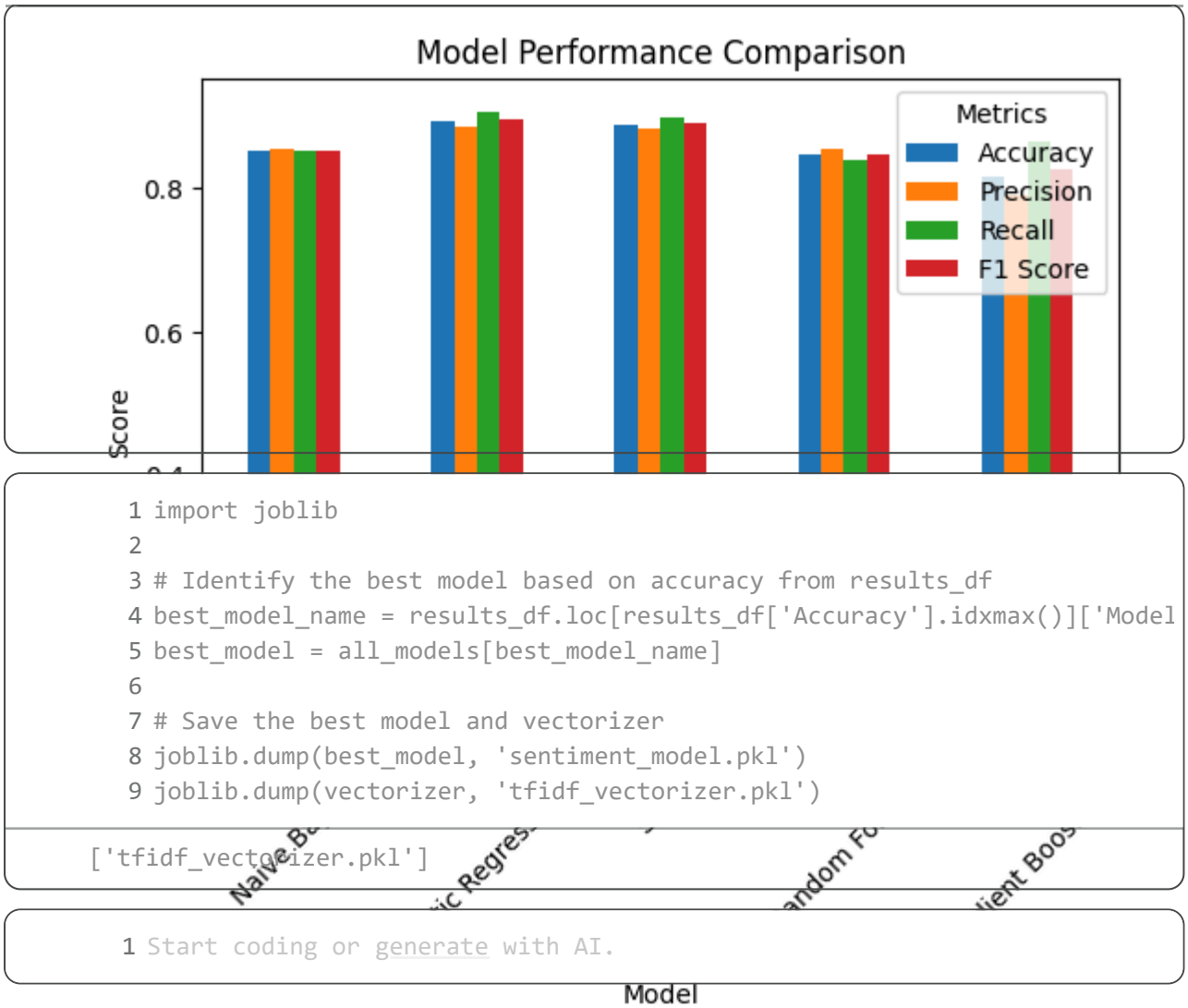

```
from sklearn.metrics import precision_score, recall_score, f1_score

results=[]
for name, model_instance in all_models.items():
  model_instance.fit(X_train,y_train)
  y_pred=model_instance.predict(X_test)
  accuracy=accuracy_score(y_test,y_pred)
  precision=precision_score(y_test,y_pred, pos_label='positive')
  recall=recall_score(y_test,y_pred, pos_label='positive')
  f1=f1_score(y_test,y_pred, pos_label='positive')
  results.append([name,accuracy,precision,recall,f1])
```

```
results_df=pd.DataFrame(results,columns=['Model','Accuracy','Precision',
results_df
```

	Model	Accuracy	Precision	Recall	F1 Score
0	Naive Bayes	0.8526	0.854585	0.852550	0.853566
1	Logistic Regression	0.8942	0.885681	0.907124	0.896275
2	SVM	0.8897	0.884075	0.898988	0.891469
3	Random Forest	0.8481	0.856275	0.839452	0.847780
4	Gradient Boosting	0.8162	0.790103	0.865053	0.825881

Next steps: Generate code with `results_df` New interactive sheet

```
import matplotlib.pyplot as plt
import seaborn as sns
results_df.set_index('Model')[['Accuracy','Precision','Recall','F1 Score
plt.title('Model Performance Comparison')
plt.ylabel('Score')
plt.xticks(rotation=45)
plt.legend(title='Metrics')
plt.show()
```




```
import joblib

# Identify the best model based on accuracy from results_df
best_model_name = results_df.loc[results_df['Accuracy'].idxmax()]['Model
best_model = all_models[best_model_name]

# Save the best model and vectorizer
joblib.dump(best_model, 'sentiment_model.pkl')
joblib.dump(vectorizer, 'tfidf_vectorizer.pkl')
```

```
['tfidf_vectorizer.pkl']
```

Start coding or generate with AI.